

MUBS Marketplace - Setup & Deployment Guide

Overview

This is a fully functional campus marketplace built with React and Firebase. Students can sign up with their MUBS email, create listings, and browse products.

Step 1: Set Up Firebase (Database & Authentication)

1.1 Create a Firebase Project

1. Go to [Firebase Console](#)
2. Click "Create a project"
3. Project name: MUBS-Marketplace (or any name)
4. Accept terms and create

1.2 Enable Authentication

1. In Firebase Console, go to **Authentication** (left sidebar)
2. Click **Get Started**
3. Click **Email/Password**
4. Enable it, save

1.3 Create Firestore Database

1. Go to **Firestore Database** (left sidebar)
2. Click **Create Database**
3. Start in **Production mode**
4. Choose a region close to Uganda (e.g., europe-west1)
5. Create

1.4 Set Up Firestore Rules (Allow MUBS-only access)

1. In Firestore, go to **Rules** tab
2. Replace the rules with:

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    match /{document=**} {
      allow read, write: if request.auth !=
      null &&

      request.auth.token.email.matches('.*@mubs.ac.ug');
    }
  }
}
```

```
}  
}  
}
```

3. Publish

1.5 Get Your Firebase Config

1. In Firebase, click the gear icon (Project Settings)
2. Go to **General** tab
3. Scroll to "Your apps" section
4. Click "Web" to create a web app
5. Copy the config object that looks like:

```
{  
  apiKey: "...",  
  authDomain: "...",  
  projectId: "...",  
  storageBucket: "...",  
  messagingSenderId: "...",  
  appId: "..."  
}
```

Step 2: Update Firebase Config in Code

1. Open `/src/firebase.js`

2. Replace the placeholder values with your actual Firebase config
 3. Save the file
-

Step 3: Install Dependencies & Run Locally

3.1 Install Node.js (if you haven't)

- Download from nodejs.org
- Choose LTS version

3.2 Install Project Dependencies

```
npm install
```

3.3 Start Development Server

```
npm start
```

This opens the app at `http://localhost:3000`

3.4 Test Locally

1. Sign up with a test MUBS email (e.g., `test@mubs.ac.ug`)
2. Create a listing

3. Go to marketplace and see it appear
 4. Everything should work!
-

Step 4: Deploy to Vercel (Free Hosting)

4.1 Sign Up on Vercel

1. Go to vercel.com
2. Click "Sign Up"
3. Choose "Continue with GitHub" (easiest)
4. Link your GitHub account

4.2 Push Code to GitHub

1. Create a GitHub account (if you don't have one)
2. Create a new repository called `mubs-marketplace`
3. Push this code to GitHub:

```
git init
git add .
git commit -m "Initial commit: MUBS Marketplace"
git branch -M main
git remote add origin
https://github.com/YOUR_USERNAME/mubs-
```

```
marketplace.git
git push -u origin main
```

4.3 Deploy on Vercel

1. Go to vercel.com/new
2. Select your GitHub repository
3. Click "Import"
4. Under "Environment Variables", add your Firebase config:
 - Key: `REACT_APP_FIREBASE_API_KEY`
 - Value: Your Firebase API key
 - Add each config value as a separate environment variable
5. Click **Deploy**

Vercel will give you a live URL like `mubs-marketplace-xyz.vercel.app` ✓

Step 5: Set Up Database Collections (One-time)

After deployment, you need to create one sample listing to test:

1. Go to your Firebase Console
2. Go to **Firestore Database**

3. Click **Start Collection**

4. Collection ID: `listings`

5. Add a document with these fields:

```
{
  name: "Sample Item",
  description: "Test item for marketplace",
  price: 50000,
  category: "cosmetics",
  sellerId: "test-id",
  sellerName: "Test Seller",
  sellerEmail: "test@mubs.ac.ug",
  verifiedSeller: true,
  status: "active",
  createdAt: <current timestamp>,
  views: 0,
  likes: 0
}
```

6. Create a `users` collection the same way
(Firestore auto-creates when first user signs up)

Step 6: Share Your Live Link!

Your marketplace is now live! Share the link with
MUBS students:

```
https://your-project.vercel.app
```

Troubleshooting

"Email doesn't match MUBS domain"

- The Firestore rules only allow `@mubs.ac.ug` emails
- If using different MUBS email domain, update the rule

"Firebase config is invalid"

- Double-check you copied the entire config from Firebase Console
- Make sure no quotes are missing

"Listings don't appear"

- Check Firestore collections exist (`listings` and `users`)
- Check browser console for errors (F12 → Console)

"Deployment fails on Vercel"

- Make sure Node version is 18+ (check `package.json`)
 - Check all Firebase config vars are added
-

Next Features to Add

1. **Messaging System** - Buyers can message sellers directly
 2. **Payment Integration** - Connect MTN Mobile Money
 3. **Seller Verification** - Admin approval process for sellers
 4. **Image Uploads** - Let sellers upload product photos
 5. **Ratings** - Buyers rate sellers after purchase
-

File Structure

```
mubs-marketplace/  
├── public/  
│   └── index.html  
├── src/  
│   ├── pages/  
│   │   ├── Auth.js  
│   │   ├── Marketplace.js  
│   │   └── SellerDashboard.js  
│   ├── styles/  
│   │   ├── App.css  
│   │   ├── Auth.css  
│   │   ├── Marketplace.css  
│   │   └── SellerDashboard.css  
│   ├── firebase.js  
│   └── App.js
```

```
|   └─ index.js  
└─ package.json  
   └─ SETUP_GUIDE.md (this file)
```

Questions?

If you get stuck:

1. Check the error message carefully
2. Google the error + "React Firebase"
3. Ask in Stack Overflow or Dev.to communities

Good luck! 🚀